6 Mutual Funds

- **Why do asset pricing researchers study managed funds?**
  - **A test of market efficiency:** if professional managers with better information and resources cannot beat passive benchmarks, prices are (close to) efficient (Jensen, 1968).
  - **A test of asset pricing models:** evaluating a fund is *evaluating a portfolio* against a benchmark—the same alpha tests we used for anomalies. A "skilled" fund may simply be loading on an omitted factor.
  - **Delegated portfolio management:** funds are agents of investors. Flows, fees, and incentives shape what managers do and what prices look like.
  - **Smart money?** Fund holdings and trades tell us who trades anomalies (momentum, value) and who is on the right side of the trade.

6.1 Mutual Funds vs. Hedge Funds: Institutional Background

	Mutual funds	Hedge funds
Regulation	Heavily regulated (Investment Company Act of 1940)	Lightly regulated private partnerships; accredited investors only
Disclosure	Daily NAV; quarterly holdings (N-CSR/N-Q, N-PORT)	Voluntary reporting to commercial databases; 13F long equity holdings for managers >$100M
Strategies	Mostly long-only, limited leverage and short selling	Long/short, leverage, derivatives, illiquid assets, dynamic trading
Fees	Expense ratio ($\approx 1\%$ per year for active equity)	"2 and 20": management fee + incentive fee, high-water marks
Liquidity	Daily redemptions	Lock-ups, redemption notice periods, gates
Benchmark	Relative return (vs. index)	Absolute return (vs. cash)
Main data	CRSP Survivor-Bias-Free MF database; Thomson/S12 holdings; MFLINKS	TASS, HFR, CISDM, Morningstar, BarclayHedge; 13F

6.2 The Key Research Questions

1. **Do active managers add value?** Do they beat passive benchmarks before and after fees?
2. **How should we measure performance?** Returns- vs. holdings-based; selectivity vs. timing.
3. **Is performance persistent?** "Hot hands" or momentum in stocks held?
4. **Skill or luck?** With thousands of funds, some will have large $t(\alpha)$ by chance.
5. **How do investors allocate capital?** The flow–performance relation; decreasing returns to scale.
6. **What do managers actually do?** Trading styles (momentum, herding), industry concentration, active share, window dressing.
7. **Do incentives distort behavior?** Tournaments and risk shifting in response to the convex flow–performance relation.
8. **Do funds affect prices?** Herding, flow-driven fire sales, "dumb money."

6.2 A map of the mutual fund literature

Question	Key papers
Do managers add value?	Jensen (1968); Sharpe (1966); Malkiel (1995); Gruber (1996); Wermers (2000)★; Fama and French (2010); Berk and van Binsbergen (2015)★
Measuring performance	Treynor (1965); Treynor and Mazuy (1966); Henriksson and Merton (1981); Lehmann and Modest (1987); Grinblatt and Titman (1989, 1993); Elton, Gruber, Das, and Hlavka (1993); Grinblatt, Titman, and Wermers (1995); Ferson and Schadt (1996); Carhart (1997)★; Daniel, Grinblatt, Titman, and Wermers (1997); Berk and van Binsbergen (2015)★
Persistence	Grinblatt and Titman (1992); Hendricks, Patel, and Zeckhauser (1993); Brown and Goetzmann (1995); Carhart (1997)★
Luck vs. skill	Kosowski, Timmermann, Wermers, and White (2006); Barras, Scaillet, and Wermers (2010); Fama and French (2010); Harvey and Liu (2018); Andrikogiannopoulou and Papakonstantinou (2019)
Fund size and scale	Berk and Green (2004)★; Chen, Hong, Huang, and Kubik (2004); Pástor, Stambaugh, and Taylor (2015)

Papers marked ★ are covered in detail below.

6.2 A map of the mutual fund literature (continued)

Question	Key papers
Flow–performance relation	Ippolito (1992); Chevalier and Ellison (1997); Sirri and Tufano (1998)★; Del Guercio and Tkac (2002); Lynch and Musto (2003); Barber, Odean, and Zheng (2005); Huang, Wei, and Yan (2007); Spiegel and Zhang (2013); Berk and van Binsbergen (2016)★; Barber, Huang, and Odean (2016)★; Ben-David, Li, Rossi, and Song (2022)
What managers do	Grinblatt, Titman, and Wermers (1995); Wermers (1999); Chen, Jegadeesh, and Wermers (2000); Kacperczyk, Sialm, and Zheng (2005, 2008); Cohen, Frazzini, and Malloy (2008); Cremers and Petajisto (2009); Amihud and Goyenko (2013); Kacperczyk, Nieuwerburgh, and Veldkamp (2014)
Incentives and agency	Lakonishok, Shleifer, Thaler, and Vishny (1991); Brown, Harlow, and Starks (1996); Chevalier and Ellison (1997); Nanda, Wang, and Zheng (2004); Kempf, Ruenzi, and Thiele (2009); Huang, Sialm, and Zhang (2011); Schwarz (2012)
Funds and prices	Wermers (1999); Coval and Stafford (2007)★; Frazzini and Lamont (2008)★; Edmans, Goldstein, and Jiang (2012); Lou (2012); Wardlaw (2020)

6.3 Tools: Measuring Fund Performance

- **Two families.** Every measure compares the fund with a **passive alternative**: **returns-based** measures build it from factor portfolios weighted by **betas**; **holdings-based** measures build it from the **characteristics** of the stocks held.
- **Returns-based measures** (regress fund excess returns on benchmark factors):

$$\begin{aligned}
\textbf{Jensen:} \quad & r_{pt} = \alpha_p + \beta_p r_{mt} + \epsilon_{pt} \\
\textbf{TM:} \quad & r_{pt} = \alpha_p + \beta_p r_{mt} + \gamma_p r_{mt}^2 + \epsilon_{pt} \\
\textbf{HM:} \quad & r_{pt} = \alpha_p + \beta_p r_{mt} + \gamma_p \max(0, r_{mt}) + \epsilon_{pt} \\
\textbf{Carhart:} \quad & r_{pt} = \alpha_p + b_p r_{mt} + s_p SMB_t + h_p HML_t + m_p UMD_t + \epsilon_{pt}
\end{aligned}$$

  TM = Treynor and Mazuy (1966); HM = Henriksson and Merton (1981); r = excess returns; $\gamma_p > 0$ indicates **market timing**.

  - **Conditional measures** (Ferson and Schadt, 1996): a refinement letting beta vary with public information.

- **Holdings-based measures** (Daniel, Grinblatt, Titman, and Wermers, 1997; Grinblatt and Titman, 1993): use the portfolio weights $w_{j,t}$ instead of fund returns.

6.3 A sequence of five papers: each one answers a problem left by the last

Paper	Method	Key finding	Problem it leaves → next paper
Grinblatt and Titman (1993)	**Holdings**: Portfolio Change Measure $\sum_j R_{j,t}(w_{j,t} - w_{j,t-k})$; past weights are the benchmark	1.9%/yr before costs (274 funds, 1975–84); aggressive growth 3.4%	Any *mechanical* tilt toward stocks with higher expected returns (e.g., past winners) counts as skill
Grinblatt, Titman, and Wermers (1995)	**Holdings**: a measure of momentum investing, $\sum_j (w_{j,t} - w_{j,t-1}) R_{j,t-1}$, on the same funds	77% of funds buy past winners; momentum funds earn the GT performance (2.6%/yr), others do not	GT performance may just be a momentum screen: need a benchmark that *controls for* size, BM, momentum
Daniel, Grinblatt, Titman, and Wermers (1997)	**Holdings**: match each stock to one of 125 size/BM/momentum portfolios; decompose into CS, CT, AS	CS $\approx$ 0.8%/yr, about the size of fees; no characteristic timing	Holdings returns are *before* fees, trading costs and cash drag: what do investors earn net?
Carhart (1997)	**Returns**: add $PR1YR$ to FF3; survivor-bias-free *net* returns	Persistence = momentum + expenses + costs; worst funds persistently bad	Net alpha < 0 while holdings show skill: are both right?
Wermers (2000)	**Holdings** + **Returns**: DGTW on holdings, Carhart on net returns, *same funds*	Stocks held beat the market by 1.3%; costs take 1.6%; net -1%	Skill covers fees but investors earn nothing extra—why? → Berk and Green (2004)

6.3 Step 1. Grinblatt and Titman (1993): Holdings-Based Measures

- **Idea:** an informed manager **increases weights on assets before they earn high returns**: performance $= \sum_j \text{cov}(w_j, R_j)$, zero for uninformed managers.
- **Portfolio Change Measure:** use the fund's *past* weights as the expected weights ($w_{j,t}$: weight at the start of period t):

$$PCM = \frac{1}{T}\sum_{t=1}^{T}\sum_{j=1}^{N} R_{j,t}\ (w_{j,t} - w_{j,t-k}) \tag{1}$$

  A zero-cost portfolio return: **no benchmark** needed, no survivorship bias, simple t-tests.

- **Evidence:** 274 funds, 1974–1984, $k = 4$ quarters: **1.9% per year** ($t = 3.13$) before costs; aggressive growth 3.4% (Table 2).
- **Caveat ($\rightarrow$ next paper):** assumes **constant expected returns**: a *mechanical* tilt toward, say, past winners (Jegadeesh and Titman, 1993) is credited as skill.

6.3 Step 2. Grinblatt, Titman, and Wermers (1995): Are Funds Momentum Traders?

- **Question:** do funds' portfolio *changes* lean toward past winners, and does that explain the GT performance?
- **Momentum measure** (GT turned around: changes in weights × *past* returns):

$$M = \frac{1}{T}\sum_{t=1}^{T}\sum_{j=1}^{N}\left(w_{j,t} - w_{j,t-1}\right) R_{j,t-k} \tag{2}$$

  $M > 0$: the fund moved toward stocks that had just done well.

- **Evidence** (155 funds, 1975–1984): $M = 0.74\%$ per quarter; **77% of funds** are momentum investors, mostly by **buying winners** (Table 1).
- **Performance:** GT performance is **2.6%/yr** for the 119 momentum funds vs. **0.1%** for the 36 contrarians (Table 2).
- **Implication:** GT "skill" may be a **momentum screen**. A benchmark must **control for the characteristics** that predict returns → DGTW.

6.3 Step 3. Daniel, Grinblatt, Titman, and Wermers (1997): Characteristic-Based Benchmarks

- **Fix:** benchmark each *stock* with a passive portfolio of stocks with the same characteristics: match stock j to one of **125 size/BM/momentum portfolios** ($5 \times 5 \times 5$) with return $R_t^{b_{j,t-1}}$ and split the holdings return:

$$
\begin{aligned}
CS_t &= \textstyle\sum_j w_{j,t-1}\left(R_{j,t} - R_t^{b_{j,t-1}}\right) && \text{(characteristic selectivity)} \\
CT_t &= \textstyle\sum_j \left(w_{j,t-1}R_t^{b_{j,t-1}} - w_{j,t-13}R_t^{b_{j,t-13}}\right) && \text{(characteristic timing)} \\
AS_t &= \textstyle\sum_j w_{j,t-13}R_t^{b_{j,t-13}} && \text{(average style)}
\end{aligned}
$$

  $CS+CT+AS =$ gross return on the stock holdings. CS: does the manager beat stocks *like* the ones held? CT: does she tilt toward characteristics before they pay off?

- **Data:** 2,500+ funds, 1975–1994, survivorship-bias-free (ten times GT's sample, twice the period).

6.3 DGTW (1997): Characteristic-based benchmarks (continued)

- **Momentum explains GT:** the GT measure is **1.94%/yr**, but regressed on the Carhart factors it falls to **0.56%** (Table III).
- **Selectivity, no timing:** **CS = 0.77%/yr** (0.56% Carhart-adjusted), about the size of fees; aggressive-growth funds about twice that; **CT ≈ 0** (Tables II–IV).
- **Same funds, factor model:** the Carhart four-factor alpha of the fund portfolio is 0.39%/yr but *insignificant*—similar point estimate, **larger standard error** (Table IV). Characteristic matching has more **power**.
- **What it leaves open:** holdings returns are *before* fees, trading costs and cash drag, and quarterly snapshots miss interim trades. What do investors receive? → Carhart (1997) (net returns) and Wermers (2000) (both).

6.3 Characteristics-Based vs. Factor-Model-Based Performance Evaluation

	Characteristics (DGTW)	Factor model (Jensen, FF3, Carhart)
Benchmark	Stocks with the **same characteristics** as each stock held; rebuilt every period from current weights	Factor portfolios weighted by **betas** estimated from the fund's return history
Data	Quarterly holdings + stock characteristics	Monthly fund returns only
Pros	Tracks the fund's *current* style (no constant-beta assumption); characteristics forecast returns better than loadings (Daniel and Titman, 1997); higher R^2, **smaller standard errors**, more power; decomposes into selectivity, timing, style; simple	Works for **any** fund (hedge, bond) from returns alone; measures what **investors receive**, net of fees, costs and cash; monthly; alpha tied to an equilibrium model, comparable across studies; easy to add factors
Cons	Gross of fees, trading costs, cash drag; snapshots miss interim trades and window dressing; depends on the characteristics chosen; needs the 125 benchmarks	Constant betas: style shifts and timing **bias alpha**; omitted factors look like skill (momentum before 1997); noisy alphas, low power; factors may not be investable, or not what investors use

Bottom line: **the question decides the tool.** Stock-picking ability? Use holdings.

What investors earn? Use net returns. Wermers (2000) uses both.

6.4 Nine Papers and How They Build on Each Other

Thread	Paper	Question	Builds on / adds
Measuring performance	Carhart (1997)	Is fund performance persistent?	Adds momentum to FF3; persistence = momentum + costs
	Wermers (2000)	Do managers add value before and after costs?	Holdings (DGTW) + net returns (Carhart): stock picking vs. style, costs, expenses
The Berk view of skill	Berk and Green (2004)	Why no net alpha or persistence if managers are skilled?	Equilibrium: competitive flows + decreasing returns to scale drive net alpha to zero
	Berk and van Binsbergen (2015)	How should skill be measured?	Dollar value added (gross alpha × size) against investable benchmarks
	Berk and van Binsbergen (2016); Barber, Huang, and Odean (2016)	Which risk model do investors use when they judge funds?	Flows as revealed preference: sign test (BvB) and flow regressions on alpha and factor-related returns (BHO); both find the CAPM
Flows and incentives	Sirri and Tufano (1998)	How do flows respond to performance?	Piecewise-linear flow regression: convexity—winners rewarded, losers not punished; implications for risk taking (Chevalier and Ellison, 1997)
Flows and prices	Coval and Stafford (2007)	Do flow-driven trades move prices?	Fire sales: pressure and reversal; refined by Edmans, Goldstein, and Jiang (2012), Wardlaw (2020)
	Frazzini and Lamont (2008)	Do flows predict the cross-section of returns?	Multi-year flow-induced ownership: "dumb money" underperforms

6.5 Carhart (1997): Persistence in Mutual Fund Performance

- **Research question:** Does persistence in fund returns reflect **skilled managers**, or is it explained by **common factors** (especially one-year momentum), **expenses**, and **transaction costs**?
- **Prior evidence of persistence:**
  - Short-term (1–3 years): "hot hands" of Hendricks, Patel, and Zeckhauser (1993); Brown and Goetzmann (1995).
  - Long-term (5–10 years), attributed to skill: Grinblatt and Titman (1992); Elton, Gruber, Das, and Hlavka (1993).
  - Grinblatt, Titman, and Wermers (1995): momentum funds outperform before costs.
- **Data:** survivor-bias-free sample of **1,892 diversified equity funds**, 1962–1993 (about one-third had died by 1993). This data later became the CRSP mutual fund database.

6.5 Carhart (1997): The four-factor model

- Carhart adds a **one-year momentum factor** to the Fama and French (1993) model:

$$r_{it} = \alpha_i + b_i RMRF_t + s_i SMB_t + h_i HML_t + p_i PR1YR_t + e_{it} \quad (3)$$

$PR1YR$ = top 30% minus bottom 30% of stocks on 11-month returns lagged one month (also known as **UMD** or **WML**).

- Carhart is agnostic: the model can be read as a 4-risk-factor model or simply as a **performance-attribution** model.

Factor (1963–1993; Table II)	Mean (% per month)	Std. dev.	t-stat
RMRF	0.47	4.43	2.01
SMB	0.29	2.89	1.89
HML	0.46	2.59	3.42
PR1YR	0.82	3.49	4.46

- On 27 size/BM/momentum portfolios, mean absolute pricing errors fall from 0.35% (CAPM) and 0.31% (3-factor) to **0.14% per month (4-factor)**.

6.5 Carhart (1997): Methodology

1. **Persistence sorts:** each January, sort funds into deciles on prior-year return; hold the equal-weighted portfolios for one year (dead funds are kept until they disappear). Regress the monthly decile returns on the CAPM and the 4-factor model.

2. **Costs and characteristics:** monthly Fama-MacBeth regressions of fund alphas on expense ratio, turnover, size, and loads:

$$\begin{aligned} \alpha_{it} &= a_t + b_t\, x_{it} + \varepsilon_{it}, \qquad \text{where} \\ \alpha_{it} &\equiv r_{it} - \hat{b}_{i,t-1} RMRF_t - \hat{s}_{i,t-1} SMB_t - \hat{h}_{i,t-1} HML_t - \hat{p}_{i,t-1} PR1Y_t \end{aligned} \tag{4}$$

   where loadings are estimated over the **prior three years** (no look-ahead).

3. **Implied transaction costs:** assume gross 4-factor alpha is zero and back out the cost per round-trip trade from alpha, expenses and turnover.

4. **Longer horizons:** sort on 2- to 5-year past returns and on past 3-year 4-factor alphas; track post-ranking returns for 5 years.

6.5 Carhart (1997): Findings—persistence is (mostly) momentum

- **Raw returns persist:** decile 1 (past winners) earns 0.68% per month, decile 10 (past losers) 0.01%: a spread of **0.67% per month ($t = 4.68$), about 8% per year**. CAPM alphas reproduce the raw spread (Table III).
- **The 4-factor model explains most of it:**
  - PR1YR loadings fall almost monotonically from 0.29 (decile 1) to −0.09 (decile 10); the 1–10 spread loads 0.38 on PR1YR ($t = 10.07$; Table III).
  - Momentum alone accounts for about 31 bp of the 67 bp monthly spread ($\approx 0.38 \times 0.82$).
  - The remaining 4-factor alpha spread (about 0.3% per month) is concentrated in the **bottom decile** (Table III).
- **"Momentum funds" do not outperform:** top funds *accidentally* hold last year's winners rather than follow momentum strategies (contrast with GTW 1995).

6.5 Carhart (1997): Findings—costs reduce performance

Fama-MacBeth regressions (Table V) of 4-factor alphas on	Coefficient	t-stat
Expense ratio	−1.54	−5.99
Turnover	−0.95	−2.36
ln(TNA)	−0.05	−0.66
Maximum load	−0.11	−3.55

- **Every 1% of expenses reduces alpha by more than 1%** (about 1.54%).
- Turnover reduces alpha by about **95 bp per round-trip trade**; load funds underperform no-load funds by about 80 bp per year.
- **Decomposing the 8% per year decile 1–10 spread:** size, BM and momentum explain 4.6%; expense ratios 0.7%; transaction costs 1.0% (Figure 3).
- The residual is concentrated in decile 10: the worst funds underperform by about *twice* their reported costs. Implied trading costs for decile 10 (354 bp per round trip) far exceed the cross-sectional estimate.

6.5 Carhart (1997): Conclusions

- **Sorting on past 3-year 4-factor alphas:** decile 1 has an insignificant post-ranking alpha of 0.02% per month; decile 10 has −0.43% ($t = -5.89$). Even after costs, decile 10 trails decile 9 by about 0.17–0.18% per month (Table VI).
- **Main conclusions:**
  - Persistence is explained by **common factors (momentum)**, **expenses** and **transaction costs**—"only very slight evidence" of skill.
  - The one anomaly: persistent underperformance of the worst funds.

6.6 Wermers (2000): Decomposing Fund Performance

- **The puzzle:** net-return studies find that active funds **underperform** (Carhart, 1997), while holdings studies find **stock-picking talent** before costs (Daniel, Grinblatt, Titman, and Wermers, 1997; Grinblatt and Titman, 1993).
- **Question:** reconcile the two. Where does the gap between the return on a fund's **stock holdings** and its **net return** go?
- **Data:** CDA quarterly holdings merged with the CRSP mutual fund database (net returns, expenses, turnover, % in stocks): **1,788 funds**, 1975–1994, free of survivorship bias.

6.6 Wermers (2000): The decomposition

- Using quarterly weights $w_{j,t-1}$ and DGTW benchmark returns $R_t^{b_{j,t-k}}$ (125 size/BM/momentum portfolios matched at $t-k$):

$$
\begin{aligned}
CS_t &= \sum_j w_{j,t-1}\left(R_{j,t} - R_t^{b_{j,t-1}}\right) \qquad \text{(stock picking)} \\
CT_t &= \sum_j \left(w_{j,t-1}R_t^{b_{j,t-1}} - w_{j,t-5}R_t^{b_{j,t-5}}\right) \qquad \text{(style timing)} \\
AS_t &= \sum_j w_{j,t-5}R_t^{b_{j,t-5}} \qquad \text{(average style)}
\end{aligned}
$$

- The gap from gross to net is split as:

$$R^{gross} - R^{net} \approx \text{expenses} + \text{trading costs} + \text{drag from cash and bonds}$$

- **Trading costs** are imputed from quarterly changes in holdings, using the Keim and Madhavan (1997) cost regressions (size of trade, market cap, price, Nasdaq dummy), scaled by year.
- **Net-return benchmark:** Carhart four-factor alpha.

6.6 Wermers (2000): Holdings returns vs. net returns

Table II. Raw returns, % per year, 1975–1994; TNA = TNA-weighted, EW = equal-weighted

			Gross (stock holdings)		Net (to investors)	
Period	S&P 500	CRSP VW	TNA	EW	TNA	EW
1975–79	15.8	18.8	20.2	22.8	16.8	18.2
1980–84	15.5	15.1	16.3	17.3	14.6	15.0
1985–89	20.8	19.0	20.2	19.5	17.4	16.4
1990–94	9.3	9.5	10.8	11.3	9.7	10.1
1975–94	**15.4**	**15.6**	**16.9**	**17.7**	**14.6**	**14.9**

- **The stocks funds hold beat the market** by **1.3%/yr** (16.9% vs. 15.6%); **investors' net return trails it** by **1.0%/yr**. The gap is 2.3%/yr.
- The gap is largest in strong stock markets (1975: 38.1% gross vs. 30.9% net) and reverses in weak ones (1977)—a sign of **cash and bonds** held for flows.

6.6 Wermers (2000): Adjusting for characteristics

Table III. Holdings: CS measure (DGTW-adjusted). Net returns: Carhart 4-factor alpha. % per year, t-stats in parentheses

	Stock holdings: CS		Net returns: Carhart α	
Period	TNA	EW	TNA	EW
1975–79	1.17 (1.78)	1.62 (2.28)	−0.81 (−1.04)	−0.77 (−0.91)
1980–84	0.71 (1.18)	1.00 (1.51)	−1.33 (−1.35)	−1.14 (−1.12)
1985–89	0.36 (1.15)	0.59 (1.35)	−0.73 (−1.21)	−1.07 (−2.01)
1990–94	0.61 (1.19)	0.82 (2.77)	−0.71 (−1.32)	−0.47 (−0.98)
1975–94	**0.71 (2.79)**	**1.01 (3.76)**	**−1.16 (−2.96)**	**−1.15 (−2.93)**

- **Same funds, opposite signs:** holdings beat their characteristic benchmarks by **+0.71%/yr**; net returns have a Carhart alpha of **−1.16%/yr**. Both are significant.
- EW > TNA: **small funds pick stocks better**. Vanguard Index 500's Carhart alpha is −0.29%, so the average fund trails it by 0.87%/yr.

6.6 Wermers (2000): Where do the returns go?

Table V, Panel A. TNA-weighted averages, % per year, 1976–1994

Period	Gross	CS	CT	AS	Expenses	Trading costs	Net	Turnover
1976–79	15.7	1.46	0.26	13.4	0.67	1.04	13.3	35%
1980–84	16.3	0.71	0.31	15.0	0.73	1.01	14.6	59%
1985–89	20.2	0.36	−0.14	20.0	0.82	0.64	17.4	73%
1990–94	10.8	0.61	−0.31	10.5	0.93	0.48	9.7	70%
1976–94	**15.8**	**0.75**	**0.02**	**14.8**	**0.79**	**0.80**	**13.8**	**61%**

- **Where the gross return comes from:** Gross = CS + CT + AS: 15.8 = 0.75+0.02+14.8. The stocks held beat the CRSP VW index (14.5%) by 1.3%/yr, of which **0.75% is stock picking** (CS) and essentially **none is style timing** (CT ≈ 0); the rest is the style tilt (AS).
- **Where it goes:** Net = Gross − expenses − trading costs − cash/bond drag: 13.8 = 15.8 − 0.79 − 0.80 − 0.4. Costs take **1.6%/yr**—more than the 1.3% the managers add.

6.6 Wermers (2000): The main finding—following the money

From the market return to investors' net return, 1976–94 (Table V, % per year, TNA-weighted)

Passive alternative: the market (CRSP value-weighted index)	14.5
+ style: funds hold characteristics with higher returns†	+0.53
+ characteristic timing (CT)	+0.02
+ stock picking within characteristics (CS)	+0.75
Gross return on stock holdings	**15.8**
− expense ratio	−0.79
− trading costs	−0.80
− lower returns on cash and bonds†	−0.41
Net return to investors	**13.8**

†Residuals: what is left after the measured components.

- Managers add **1.3%**; costs take **1.6%**; cash and bonds take about **0.4%** ⇒ investors trail the market by **0.7%**.

6.6 Wermers (2000): What the main finding means

- **Managers have skill:** their stocks beat the market by 1.3%/yr (0.75% pure stock picking), roughly covering their 1.6% of costs—as Grossman and Stiglitz (1980) predict.
- **Investors still underperform** by 0.7%/yr: costs exceed the 1.3% gain by 0.3%, and cash and bonds cost another 0.4%. Holdings studies (before costs) and net-return studies (after costs) are **both right**.

6.7 From Measuring Alpha to Measuring Skill: The Berk Turn

- **What the measurement literature established** (GT → GTW → DGTW → Carhart → Wermers):
  - **Before costs**, managers' stocks beat characteristic- or factor-matched benchmarks by roughly **the size of their fees** (CS ≈ 0.7–0.8%/yr).
  - **After costs**, net alpha is about **zero or negative** and does **not persist** once momentum is controlled for.
  - Yet investors **chase past returns** (Chevalier and Ellison, 1997; Sirri and Tufano, 1998) and managers are highly paid.
- **The standard reading:** managers have no skill; investors who chase returns are irrational.
- **Berk's different angle:** these are **equilibrium outcomes**, and **alpha is the wrong object**. Every alpha above is a *return per dollar*. If capital flows competitively to skilled managers and skill has decreasing returns to scale, funds grow until the return per dollar to investors is zero: **zero net alpha is what skill looks like in equilibrium**.

6.7 The Berk turn: three papers, one idea—look at quantities, not returns

- **Berk and Green (2004):** a rational model with competitive capital, decreasing returns to scale and learning. Flows, not prices, restore equilibrium; net alpha $\to 0$ for every fund; performance chasing is rational; rents go to managers through **fund size**.
- **Berk and van Binsbergen (2015):** so measure skill as a **quantity**—gross alpha $\times$ dollars managed (**value added**)—against the investor's **investable** alternative (Vanguard index funds), not statistical factors. This revisits the benchmark debate from the investor's side.
- **Berk and van Binsbergen (2016) and Barber, Huang, and Odean (2016):** if flows restore equilibrium, flows reveal which benchmark *investors* use. Test asset pricing models with **flows instead of returns**.
- **The common thread:** replace "is the return per dollar abnormal?" with "where does the money go, and what does that reveal?" **Fund size and flows** carry the information about skill.

6.8 Berk and Green (2004): Flows and Performance in Rational Markets

- **The puzzles** (from the measurement literature just reviewed):
  1. On average, active managers **do not outperform** passive benchmarks net of costs (Carhart, 1997; Wermers, 2000).
  2. Performance is largely **not persistent** (Carhart, 1997).
  3. Yet investors **chase performance**: flows respond strongly (and convexly) to past returns (Chevalier and Ellison, 1997; Sirri and Tufano, 1998).
  4. And managers are **highly paid**.
- **Standard interpretation:** managers have no skill and investors are irrational.
- **Berk and Green's question:** What does a **fully rational** model—no moral hazard, no asymmetric information—imply for flows, performance, persistence, and survival?
- **Key idea:** like in corporate finance, capital is supplied **competitively**, so rents go to the scarce resource: managerial talent. In open-end funds the equilibrating variable is **quantity (fund flows), not price**.

6.8 Berk-Green (2004): The model

- **Skill:** the return on the first dollar of active management (over the benchmark, before costs) is $R_t = \alpha + \epsilon_t$, $\epsilon_t \sim N(0, \sigma^2)$; nobody knows α, prior $\alpha \sim N(\phi_0, \eta^2)$.
- **Decreasing returns to scale:** running q dollars actively costs $C(q)$, $C' > 0$, $C'' > 0$; the fee is f per dollar. Investors in a fund of size q_{t-1} earn

$$r_t = R_t - c(q_{t-1}), \qquad c(q) \equiv \frac{C(q)}{q} + f \quad \text{(increasing in } q\text{)}.$$

- **Learning:** the estimate of skill $\phi_t \equiv E_t[R_{t+1}]$ is updated by Bayes' rule, with a weight on R_t that **falls with fund age** t ($\omega = 1/\sigma^2$, $\gamma = 1/\eta^2$):

$$\phi_t = \frac{\gamma + (t-1)\omega}{\gamma + t\omega}\,\phi_{t-1} + \frac{\omega}{\gamma + t\omega}\,R_t.$$

- **Competitive capital:** money flows in while $E_t(r_{t+1}) > 0$ and out while $E_t(r_{t+1}) < 0$, so in equilibrium

$$E_t(r_{t+1}) = 0 \quad \Longleftrightarrow \quad \phi_t = c(q_t)$$

The fund grows until perceived skill just covers fees and the cost of scale.

6.8 Berk-Green (2004): Key predictions

1. **Zero expected net alpha for every fund**, skilled or not: $E_t(r_{t+1}) = 0$ is the equilibrium condition. Skill is real, but its rents go to the **manager** through fund size (fee $\times$ AUM), not to investors.
2. **No persistence in net performance**, though α is persistent: a good year raises ϕ_t, the fund grows, and the extra money dilutes alpha back to zero.
3. **Flows chase performance, rationally.** Combining the equilibrium condition with the learning rule gives the flow–performance relation

$$c(q_t) = c(q_{t-1}) + \frac{\omega}{\gamma + t\omega} r_t : \tag{5}$$

   $r_t > 0 \Rightarrow$ inflow, $r_t < 0 \Rightarrow$ outflow; **stronger for young funds** (the weight falls with t) and, with quadratic costs, **convex** in r_t.
4. **Rational closet indexing:** with a fixed fee the manager runs only the optimal amount actively and indexes the rest; new money after good performance is indexed, so fund **volatility falls**—no agency story needed.
5. **Exit:** a fund closes when ϕ_t falls below a threshold, so poor past returns predict disappearance.

6.8 Berk-Green (2004): Does the model fit the facts?

Parameter	Symbol	Value	Source
Fee	f	1.5%	Average expense ratio plus amortized loads
Return volatility	σ	20%	Historical portfolio volatility
Exit threshold	$\underline{\phi}$	3%	$= 2f$
Prior mean of skill	ϕ_0	6.5%	Fitted
Prior std. dev. of skill	η	6%	Fitted ($\gamma = 277$)

- **Context:** the model is not estimated but **calibrated**: the prior mean and dispersion of skill are chosen to match **fund survival rates** (CRSP, 1969–1999) and the **flow–performance curve** of Chevalier and Ellison (1997).
- **What the data say:** the model reproduces the convex flow–performance curve and short-horizon survival rates (Figures 2 and 3). To fit them, **about 80% of managers must have skill above the 1.5% fee** (Figure 4)—the evidence usually read as "no skill" is consistent with *substantial* skill.
- **Misses:** long-run survival in the data falls faster than in the model (Figure 2).

6.8 Berk-Green (2004): Implications and legacy

- **Lack of persistence and zero net alpha do NOT imply that managers lack skill.**
  - Tests of skill based on net alpha or alpha persistence are tests of *investor* returns, not of *manager* ability.
- **Measuring skill as value added:** Berk and van Binsbergen (2015) measure skill as **dollar value added** (gross alpha × fund size) and find it is positive and persistent.
- **Decreasing returns to scale:** Chen, Hong, Huang, and Kubik (2004) find fund size erodes performance; Pástor, Stambaugh, and Taylor (2015) find stronger **industry-level** than fund-level decreasing returns.
- **Which model do investors use?** Berk and van Binsbergen (2016) and Barber, Huang, and Odean (2016) use flows to infer investors' risk model—flows respond most to CAPM alpha.
- **Critiques:** perfectly elastic and rational capital (no search costs, sticky money); calibration rather than estimation; does not explain the persistent underperformance of the worst funds (Carhart, 1997).

6.9 Berk and van Binsbergen (2015): Measuring Skill

- **Question:** if Berk and Green (2004) is right, net alpha is zero in equilibrium. **How should skill be measured?**
- **Alpha is a return, not a value.** 1% gross alpha on a \$10B fund adds more value than 10% on a \$1M fund.
  - Peter Lynch (Magellan): about 2%/month gross alpha on \$40M in his first five years; 0.2%/month on more than \$10B in his last five. His value added rose from under \$1M to over \$20M per month.
- **Value added** of fund i in month t, with lagged AUM $q_{i,t-1}$ (real dollars) and benchmark return R^B_{it}:

$$V_{it} \equiv q_{i,t-1}\left(R^g_{it} - R^B_{it}\right) = \underbrace{q_{i,t-1} f_{i,t-1}}_{\text{dollar fees}} + \underbrace{q_{i,t-1}\left(R^n_{it} - R^B_{it}\right)}_{\text{to investors}} \tag{6}$$

Skill is $S_i \equiv E[V_{it}]$, estimated by the time-series average of V_{it}.

6.9 Berk-van Binsbergen (2015): Why alpha fails, and the benchmark

- **Proposition 1 (in words):** in the Berk–Green equilibrium a manager runs the optimal amount actively and indexes the rest, so
  - **net alpha** is zero for everyone—it never measures skill;
  - **gross alpha** equals the fee, which is spread over a fund size that rises with skill—it ranks managers correctly only if all funds have the same size;
  - **value added** = gross alpha × size is the manager's rent and always measures skill.
- **Benchmark = the investor's alternative:** the fund's closest combination of the 11 **Vanguard index funds** available at each date (S&P 500 from 1976, then extended market, small-cap, value, international, ...).
- **Why Vanguard rather than the FFC factors?** The benchmark should be what investors **could actually have bought**; the FFC long–short factors are not tradable. Against Vanguard, SMB, HML and UMD have alphas of 22, 35 and 70 bp/month (Table 2): FFC credits investors with returns they could not earn.

6.9 Berk-van Binsbergen (2015): Average value added

Value added $V_{it} = q_{i,t-1}(R^g_{it} - R^B_{it})$, Y2000 \$ million per month; 5,974 active funds, 1977–2011 (Table 3)

	Vanguard benchmark	FFC factors
Time-weighted mean	0.27 ($t = 5.74$)	0.25 ($t = 3.94$)
Simple (ex ante) mean	0.14 ($t = 4.57$)	0.10 ($t = 3.43$)
Median	−0.02	−0.03
90th / 99th percentile	0.75 / 7.82	0.70 / 6.76
% of funds with $\hat{S}_i < 0$	57%	60%

- **How the numbers are built:** for each fund, $\hat{S}_i$ = time-series average of monthly V_{it}; the **time-weighted mean** across funds weights each $\hat{S}_i$ by the number of months the fund exists (the cross-sectional mean of all fund-months); the **simple mean** weights funds equally.
- The average fund adds **\$3.2 million per year** (= 0.27 \$M/month × 12). The median fund destroys value (−0.02), but the skilled managers run most of the money.

6.9 Berk-van Binsbergen (2015): Who gets the value added? Investors vs. managers

Net alpha of a dollar invested each month across all funds, bp/month (t-statistics in parentheses; Table 6)

Strategy	Vanguard benchmark	FFC factors
Equal-weighted across funds	2.74 (0.73)	−3.88 (−1.40)
Value-weighted across funds	−0.95 (−0.31)	−5.88 (−2.35)

- **Investors earn nothing extra:** against the Vanguard benchmark, net alpha is **not different from zero**. The value added accrues to **managers** as fees, as Berk–Green predict. (Against FFC it is negative—but FFC was not an investable alternative.)

6.9 Berk-van Binsbergen (2015): Is skill persistent?

- **Test:** sort funds into deciles on the **skill ratio** (the t-statistic of past value added); measure value added *out of sample* over the next 3 to 10 years. Under the null of no skill, "top decile beats bottom decile" in a given month is a coin flip, so the count is Binomial$(n, 1/2)$—a test robust to fat tails and cross-correlation.
- **Result:** the top decile adds about \$1.1–3.4M per month out of sample and beats the bottom decile 54–62% of the time; the null is rejected at almost all horizons (Table 4, Figure 3). **Skill persists for up to 10 years.**
- **Investors recognize it:** fees are similar across skill deciles but AUM is not, so **compensation** (fee $\times$ AUM) rises steeply with skill, and current compensation predicts future value added (Table 5)—skill shows up in fund **size**, not in alpha.

6.10 Berk and van Binsbergen (2016): Revealed Preference

- **Question:** which asset pricing model do investors actually use? Test models with **quantities (flows)** instead of returns.
- **Logic:** every neoclassical model assumes investors compete away positive-NPV opportunities. In mutual funds the price is fixed at NAV, so equilibrium is restored by **flows** (Berk and Green, 2004):
  - outperformance under the *true* model ($\varepsilon_{it} > 0$) is good news about skill $\Rightarrow$ inflows; underperformance $\Rightarrow$ outflows.
- **Test:** use only the **signs** of flows F_{it} and of model-implied outperformance $\varepsilon_{it} = R^n_{it} - R^B_{it}$:

$$\beta_{F\varepsilon} = \frac{\text{cov}(\phi(F_{it}), \phi(\varepsilon_{it}))}{\text{var}(\phi(\varepsilon_{it}))} > 0 \tag{7}$$

  $(\beta_{F\varepsilon} + 1)/2$ is the average probability that the sign of the flow matches the sign of ε (50% = no relation).

- The **true model maximizes** $\beta_{F\varepsilon}$; models are ranked pairwise with double-clustered t-statistics.

6.10 Berk-van Binsbergen (2016): Which model wins?

Sign-agreement probability $(\beta_{F\varepsilon} + 1)/2$, %; 4,275 funds, 1977–2011 (Table 2)

Model	3 months	1 year	4 years
CAPM (CRSP VW)	63.6	63.4	63.4
Fama–French 3-factor	63.1	63.1	62.4
Fama–French–Carhart	63.3	63.1	62.4
Return in excess of market	62.1	61.2	61.8
Raw return (no model)	58.6	57.7	61.2
Consumption CAPM	58.3	57.4	61.2
Habit (Campbell–Cochrane)	58.3	57.4	61.0

- The **CAPM ranks first** at all horizons except two years (Table 3). It significantly beats "excess of market" at all horizons and "no model" at horizons under two years (Table 4)—**investors use beta**.
- FF and FFC **never significantly beat the CAPM**; dynamic equilibrium models do significantly worse at horizons up to one year (Table 4).

6.10 Berk-van Binsbergen (2016): Interpretation

- **Interpretation:** the extra factors in FF and FFC **do not measure risk investors care about**; their premia may reflect non-risk factors. No model explains more than about 64% of flow signs (Table 2).
- **Why only signs?** The sign test needs no assumption about the *shape* of the flow–performance relation (which is convex) or about the scale of flows; the cost is that it throws away the magnitudes of both flows and outperformance.
- **Critiques:** (i) the test identifies the model *investors* use, not necessarily the *true* model—investors may follow heuristics; (ii) the CAPM's edge over FF/FFC is small (63.6 vs. 63.1%); (iii) raw returns and "excess of market" are close behind, so part of the result is "investors adjust for beta," not "investors believe the CAPM." Barber, Huang, and Odean (2016) address (ii) and (iii) with a more standard design.

6.11 Barber, Huang, and Odean (2016): Which Factors Matter to Investors?

- **Question:** which components of past returns do investors **discount as "not skill"**? A sophisticated investor strips out *every* factor that explains cross-sectional returns, priced or not.
- **Same idea as BvB (2016), standard tools: flow–performance regressions in the Sirri–Tufano tradition**, with the past return **decomposed** into alpha and factor-related pieces—interpretable magnitudes, easy controls; the specification most later papers use.
- **Two designs:**
  1. **Horse race:** which model's alpha best predicts flows? MAR, CAPM, FF3, Carhart 4F, 7F (+ three industry factors), 9F (+ profitability, investment).
  2. **Return decomposition:** flows on alpha *and* each factor-related return: a factor investors use gets a **zero** coefficient; one they ignore gets alpha's coefficient.
- **Data: about 4,000 active U.S. equity funds**, monthly, 1996–2011; $F_{pt} = TNA_{pt}/TNA_{p,t-1} - (1 + R_{pt})$.

6.11 BHO (2016): Methodology (1)—rolling loadings and the return decomposition

1. **Rolling factor loadings.** For fund p in month t, estimate the model on the prior 60 months; e.g., the 7-factor model:

$$\begin{aligned} R_{p\tau} - R_{f\tau} \;=\; & \alpha_{pt} + \beta_{pt} RMRF_\tau + s_{pt} SMB_\tau + h_{pt} HML_\tau \\ & + m_{pt} UMD_\tau + \textstyle\sum_{k=1}^{3} i_{pt}^k IND_\tau^k + e_{p\tau} \end{aligned}$$

2. **Month-t realized alpha and return components** (out of sample, with lagged loadings):

$$\begin{aligned} R_{pt} - R_{ft} \;=\; & \underbrace{\hat{\alpha}_{pt}}_{ALPHA} + \underbrace{\hat{\beta}_{pt} RMRF_t}_{MKTRET} + \underbrace{\hat{s}_{pt} SMB_t}_{SIZRET} + \underbrace{\hat{h}_{pt} HML_t}_{VALRET} \\ & + \underbrace{\hat{m}_{pt} UMD_t}_{MOMRET} + \textstyle\sum_k \underbrace{\hat{i}_{pt}^k IND_t^k}_{INDRET_k} \end{aligned}$$

3. **Horizon.** Flows respond to returns over the prior **18 months** with **exponentially decaying** weights (AIC; Figure 1); each component is averaged over $t-1, \ldots, t-18$ with weights $\propto e^{-\lambda s}$.

6.11 BHO (2016): Methodology (2)—the panel regression and how to read it

- **Panel regression** (month fixed effects μ_t; standard errors double-clustered by fund and month):

$$\begin{aligned} F_{pt} \;=\; & b_0 + b_1\, ALPHA_{pt} + b_2\, MKTRET_{pt} + b_3\, SIZRET_{pt} \\ & + b_4\, VALRET_{pt} + b_5\, MOMRET_{pt} + \textstyle\sum_{k=1}^{3} b_{5+k}\, INDRET_{k,pt} \\ & + \gamma' X_{pt} + \mu_t + e_{pt} \end{aligned}$$

  Controls X: expense ratio, no-load dummy, return volatility, log size, log age, lagged flows.

- **Reading the coefficients:**
  - Sophisticated (7-factor) investors: $b_1 > 0$, $b_2 = \cdots = b_8 = 0$. Naive (raw-return) investors: all b equal.
  - CAPM investors: $b_2 = 0$ and $b_1 = b_3 = \cdots = b_8$—style returns treated as alpha.

6.11 BHO (2016): Horse race—CAPM alpha predicts flows best

Flows on six measures jointly (Table 3) and pairwise decile-rank horse races (Table 4, Panel A)

Table 3: all six in one regression

Measure	Coef. (raw)	Pctile rank
CAPM alpha	0.474***	1.357***
Market-adjusted return	0.221***	0.852***
Three-factor alpha	0.186***	0.361***
Four-factor alpha	−0.072	0.039
Seven-factor alpha	0.071	0.285***
Nine-factor alpha	0.005	−0.177*

Table 4A: CAPM vs. each rival

Rival	$\sum$(coef. diff.)	% > 0
MAR	7.41 ($t = 3.5$)	78%
3-factor	22.94 ($t = 10.6$)	100%
4-factor	27.50 ($t = 13.3$)	100%
7-factor	28.52 ($t = 14.7$)	100%
9-factor	33.03 ($t = 17.6$)	100%

- **Table 3:** 1 pp more CAPM alpha brings 0.47 pp more monthly flow—twice the effect of the market-adjusted return; multifactor alphas add little once the CAPM alpha is in.
- **Table 4 (nonparametric):** sort funds into deciles on two models' alphas; dummies for (CAPM decile i, rival decile j); compare b_{ij} with b_{ji} over all 45 pairs. The CAPM wins every comparison. Ranking: **CAPM** $\succ$ **MAR** $\succ$ **3F** $\succ$ **4F** $\succ$ **7F** $\succ$ **9F**.

6.11 BHO (2016): Return decomposition—investors adjust for beta, not for style

Flows on the components of the prior 18 months' return (Table 5, column 1)

Return component	Coefficient	(s.e.)	Ratio to $ALPHA$
$ALPHA$ (7-factor)	0.884	(0.027)	1.00
$MKTRET$ (beta × market)	0.253	(0.056)	0.29
$SIZRET$	0.759	(0.054)	0.86
$VALRET$	0.665	(0.063)	0.75
$MOMRET$	1.059	(0.060)	1.20
$INDRET_1$, $INDRET_2$, $INDRET_3$	0.92, 0.70, 0.69		1.04, 0.79, 0.78

- **Investors discount market-related returns** (29% of the alpha response) but treat **size, value, momentum and industry returns largely as alpha** (75–120%). Momentum is rewarded *at least* as much as alpha.
- **Sophistication (Table 7):** broker-sold funds, high-sentiment months and less wealthy households respond **more** to factor-related returns.

6.11 BHO (2016) vs. BvB (2016): two designs, one message

	Berk and van Binsbergen (2016)	**Barber, Huang, and Odean (2016)**
Dependent variable	Sign of flow	Percentage flow
Performance	Sign of outperformance; full-sample betas vs. Vanguard funds	Rolling 60-month loadings; return **decomposed**; 18-month decay weights
Test	Sign agreement $\beta_{F\varepsilon}$; pairwise model tests	Panel regressions with controls/FE; decile horse race; coefficient ratios
Strength	Model-free about the shape of the flow relation	Interpretable magnitudes: *how much* of each factor is discounted; subsamples
Finding	CAPM ranks first; FF/FFC never beat it	CAPM alpha wins; style returns treated as alpha

- **Interpretation differs:** BvB: evidence on the **true asset pricing model**. BHO: evidence on **how investors evaluate managers**—a heuristic that confuses style with skill, strongest among less sophisticated investors.

6.12 Sirri and Tufano (1998): The Flow–Performance Relation Is Convex

- **Research question:** How do flows into equity funds respond to past performance? **Symmetrically**—rewarding winners *and* punishing losers—or not? Secondary: fees, risk, and search costs.
- **Why it matters:** fees are a percentage of assets, so the **shape** of the flow–performance relation is the fund company's implicit incentive contract (Brown, Harlow, and Starks, 1996; Chevalier and Ellison, 1997). A convex relation is a **call option** on relative performance.
- **Framing:** buying a fund is more like buying a **car** than picking a portfolio—many brands, advertising, salespeople, costly information.
- **Related evidence:** Ippolito (1992): growth responds more to positive than to negative excess returns; Chevalier and Ellison (1997): a semiparametric (kernel) estimate of the same curve, also convex.
- **What the paper is remembered for:** the **piecewise-linear flow regression**, still the standard specification for fund flows.

6.12 Sirri-Tufano (1998): Predictions and data

- **Benchmark prediction (costless search):** investors following the academic evidence would **leave the worst funds** (poor performance persists, Carhart, 1997), ignore good past performance (it does not persist), and avoid **risk** and **fees**.
- **Data:** ICDI, 1971–1990: **690 U.S. equity funds** (aggressive growth, long-term growth, growth and income), 288 families; dead funds only for 1987–1990.
- **Dependent variable:** annual net flow as a fraction of beginning assets (flows at year end; new money earns the fund's return—the standard construction ever since):

$$Flow_{i,t} = \frac{TNA_{i,t} - TNA_{i,t-1}(1 + R_{i,t})}{TNA_{i,t-1}}$$

- **Controls:** log lagged TNA; flows to the fund's objective; std. dev. of prior-year monthly returns; **total fees** = expense ratio + load/7.
- **Estimation:** **Fama–MacBeth**—a cross-sectional regression each year; standard errors from the time series of coefficients (robust to cross-sectional correlation within a year).

6.12 Sirri-Tufano (1998): The key method—a piecewise-linear flow regression

- **Performance variable:** the fund's **fractional rank** $Rank_{i,t-1} \in [0,1]$ of the prior-year raw return *within its objective category* ($0 =$ worst, $1 =$ best); ranks are comparable across years.
- **Piecewise-linear (linear spline) specification:** split the rank into quintile segments, each with **its own slope**:

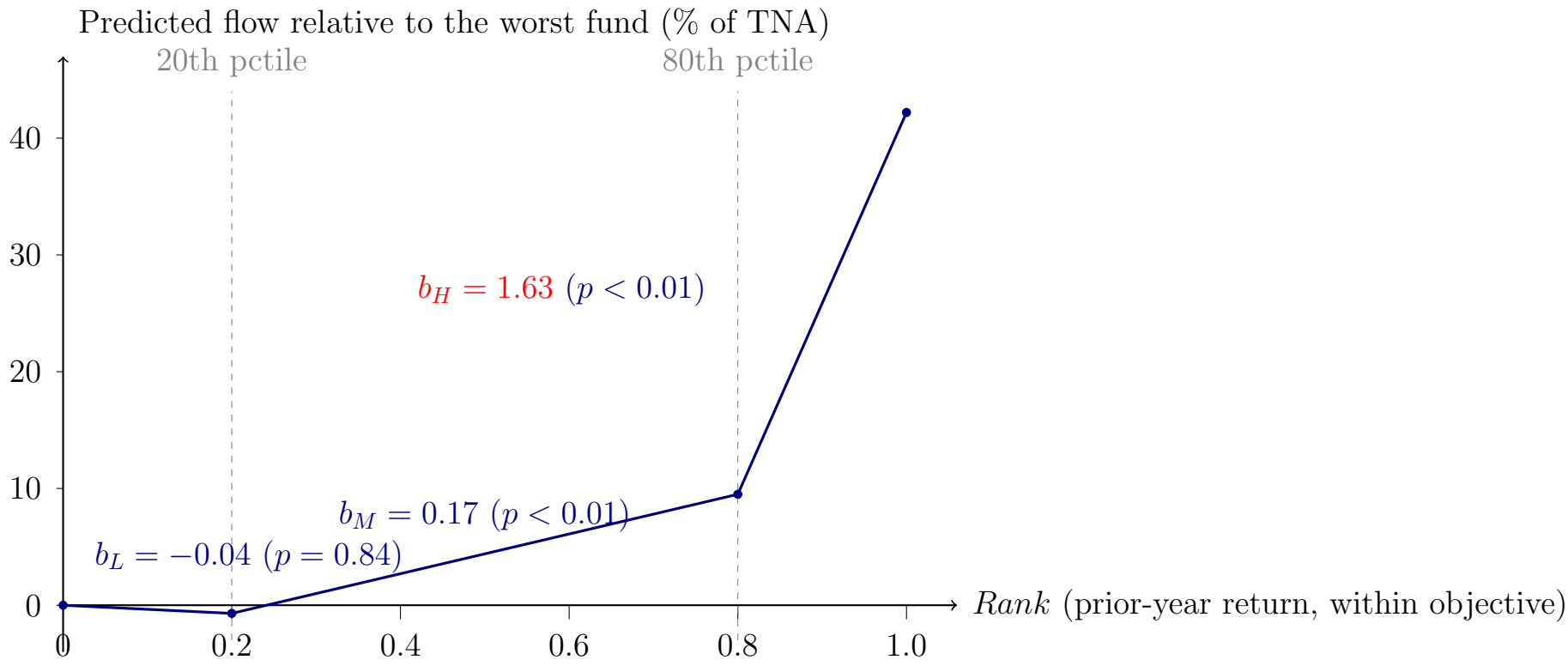
$$\begin{aligned} LOW_i &= \min(Rank_i,\, 0.2) \\ MID_i &= \min(Rank_i - LOW_i,\, 0.6) \\ HIGH_i &= Rank_i - LOW_i - MID_i \\ Flow_{i,t} &= a + b_L\, LOW_i + b_M\, MID_i + b_H\, HIGH_i + c'X_i + e_{i,t} \end{aligned}$$

- **How to read it** (all regressors measured in year $t-1$): a fund at rank 0.9 has $LOW = 0.2$, $MID = 0.6$, $HIGH = 0.1$. The fitted line is **continuous** at the kinks (0.2, 0.8); b_L, b_M, b_H are the slopes in the bottom quintile, the middle three quintiles, and the top quintile. Moving up 10 percentile points inside the top quintile adds $0.1 \times b_H$ to flows.

6.12 Sirri-Tufano (1998): Why a piecewise-linear specification?

- **It nests the linear model** ($b_L = b_M = b_H$) and gives a direct **test of nonlinearity**: $b_H > b_M > b_L$ is convexity; $b_L \approx 0$ means losers are not punished.
- **Parsimonious and interpretable:** three slopes instead of quintile or decile dummies (which lose the within-segment slope), and each slope reads as an **incentive schedule**—the marginal flow reward for one more percentile of performance in that region.
- **Flexible:** breakpoints can be moved (e.g., above/below median); the segments can be **interacted** with fund characteristics—fees, age, size, complex—to ask *who* faces the steeper option (Huang, Wei, and Yan, 2007; Sirri and Tufano, 1998); the same spline works with alphas or multi-year returns as the performance measure.
- **Alternative:** fully nonparametric (kernel) estimation (Chevalier and Ellison, 1997) lets the data pick the shape but is harder to interact and to compare across samples.

6.12 Sirri-Tufano (1998): The estimated relation is convex



- Fitted values, Table II, column 1. **Flat at the bottom, gently rising in the middle, steep at the top:** the top-quintile slope is about ten times the middle slope; the 85th → 90th percentile brings **8.4%** more money, a similar move elsewhere 0–1.4%.

6.12 Sirri-Tufano (1998): Flows chase winners, losers are not punished

Dependent variable: $Flow_{i,t}$ (Table II, Fama–MacBeth, 1971–1990)	Coefficient	p-value
Bottom quintile (LOW)	−0.035	0.843
Middle three quintiles (MID)	0.170	0.000
Top quintile ($HIGH$)	1.633	0.000
Total fees	−0.029	0.057
Std. dev. of monthly returns	−1.068	0.101
Log lagged TNA	−0.048	0.000
Flows to fund category	0.965	0.003
Adjusted R^2 / observations	14.3% / 4,098	

- **Convex:** $b_H \gg b_M > b_L \approx 0$; the linear null ($b_L = b_M = b_H$) is rejected. **Losers are not punished:** the bottom-quintile slope is zero—investors in the worst funds do not leave.

6.12 Sirri-Tufano (1998): Why convexity matters—an option and an agency problem

- **The fund company's payoff is convex in relative performance.** Revenue = fee × TNA, and next year's TNA is $TNA_t(1+Flow)$ with $Flow$ convex in this year's rank: a **top-quintile** finish brings a large inflow, a **bottom-quintile** finish costs almost nothing. The company holds a **call option** on its relative return.
- **An agency problem.** Investors want the manager to maximize *risk-adjusted* return; the company wants to maximize *expected assets*. With a convex payoff the company gains from **variance**: a gamble that moves the fund from the 50th to the 90th percentile is worth far more than the same gamble to the 10th percentile costs.
- **Predictions for behavior: tournaments** (Brown, Harlow, and Starks, 1996)—mid-year losers raise volatility, leaders lock in; **year-end risk shifting** (Chevalier and Ellison, 1997); closet indexing by leaders; window dressing (Lakonishok, Shleifer, Thaler, and Vishny, 1991).

6.12 Sirri-Tufano (1998): Where does the convexity come from? And what follows

- **Rational explanations of the convexity:** learning with decreasing returns to scale (Berk and Green, 2004); poor performance triggers manager or strategy changes, so losers are not left (Lynch and Musto, 2003); participation costs (Huang, Wei, and Yan, 2007); percentage flows overstate convexity—market-share flows are closer to linear (Spiegel and Zhang, 2013).
- **Behavioral/marketing explanations:** search costs and advertising (this paper); inert investors; attention to salient winners.
- **Later work:** retail flows are convex, pension sponsors use risk-adjusted performance and fire losers (Del Guercio and Tkac, 2002); investors respond to visible loads, not expenses (Barber, Odean, and Zheng, 2005); Morningstar ratings (Ben-David, Li, Rossi, and Song, 2022); CAPM alpha (Barber, Huang, and Odean, 2016; Berk and van Binsbergen, 2016).

⇒ **Whatever its source, if the flow curve is convex the manager's incentive problem follows.** Evidence on the response: Chevalier and Ellison (1997).

6.12 Sirri-Tufano (1998): Do managers respond to the convexity? Evidence from Chevalier and Ellison (1997)

- **The flow curve without a functional form:** kernel regression of next year's flow on this year's market-adjusted return (449 growth and growth-and-income funds, 1982–1992). **Convex**, and steeper for **young** funds: a 2-year-old fund 8 points behind the market expects a 14% outflow; 3 points behind, a small inflow; far ahead, flows rise sharply.
- **The incentive it creates:** for a fund *somewhat* behind the market in September, a 50/50 gamble in Q4 raises expected assets more than indexing does (gaining 5 points nearly eliminates the outflow; losing 5 costs little); a fund ahead should lock in. From the curve they compute each fund's **risk incentive** given its January–September return.
- **Managers act on it:** the change in portfolio risk between the September and December holdings rises with the risk incentive ($t \approx 3$), more for small funds (Table 4): funds a few points behind the market **increase** tracking-error risk at year end; funds far behind or ahead do not (Table 5).
- **Later evidence:** Brown, Harlow, and Starks (1996); Schwarz (2012); Kempf, Ruenzi, and Thiele (2009); Huang, Sialm, and Zhang (2011).

6.13 Coval and Stafford (2007): Asset Fire Sales

- **Question:** do fund flows **move stock prices**? When funds are *forced* to trade, do prices move away from fundamental value and then revert?
- **Fire sales** (Shleifer and Vishny, 1992): assets are sold below value when the natural buyers are also constrained. Here the "natural buyers" are other funds holding the same stocks; arbitrage is limited (Merton, 1987; Shleifer and Vishny, 1997).
- **Why open-end funds are a clean setting:** capital is redeemable on demand, funds hold little cash and cannot short or borrow, and both holdings and flows are observed.
- **Data:** CDA/Spectrum quarterly holdings merged with CRSP flows, 1980–2004 (142,720 fund-quarters):

$$flow_{j,t} = \frac{TNA_{j,t} - TNA_{j,t-1}(1 + R_{j,t})}{TNA_{j,t-1}}$$

- **Funds with outflows sell**: the bottom flow decile (−12.5% per quarter) reduces or eliminates 54% of its positions; the top decile expands 46% of its existing positions (Table 2, Panel B).

6.13 Coval-Stafford (2007): Measuring flow-driven pressure

- **As defined in the paper (equation (4)):** for stock i in quarter t, with $\Delta Holdings_{j,i,t}$ the change in shares held by fund j,

$$PRESSURE_1_{i,t} = \frac{\sum_j \max(0, \Delta Holdings_{j,i,t}) \big|_{flow_{j,t} > Percentile(90th)} - \sum_j \max(0, -\Delta Holdings_{j,i,t}) \big|_{flow_{j,t} < Percentile(10th)}}{AvgVolume_{i,t-12:t-6}}$$

- **In words:** shares of stock i **bought** by funds with **extreme inflows**, minus shares **sold** by funds with **extreme outflows**, as a fraction of the stock's normal trading volume:

$$PRESSURE_{i,t} = \frac{\text{Buys}_{i,t}\big|_{\text{inflow funds}} - \text{Sells}_{i,t}\big|_{\text{outflow funds}}}{AvgVolume_i}$$

- **Alternatives (equations (5)–(6)), used for robustness in Table 5**; results are "remarkably similar" across the three:
  - $PRESSURE_2$ weights each fund's trade by the *size* of its flow (rather than counting only funds in the extreme flow deciles).
  - $PRESSURE_3$ is $PRESSURE_1$ with *shares outstanding* instead of trading volume in the denominator.

6.13 Coval-Stafford (2007): Measuring flow-driven pressure (continued)

- **Fire sales:** stocks in the bottom decile of $PRESSURE$; **forced purchases:** top decile. (At least ten mutual fund owners required.)
- **Control group:** *unforced* sales—net selling by all funds, **not** conditioned on flows, likely informed:

$$Unforced_PRESSURE_{i,t} = \frac{\sum_j \Delta Holdings_{j,i,t}}{AvgVolume_{i,t-12:t-6}}$$

- **Tests:** event-time abnormal returns (vs. the average stock held by funds), months -12 to $+18$; calendar-time alphas of trading strategies, including **front-running** predicted flows.

6.13 Coval-Stafford (2007): Price pressure and reversal

Cumulative average abnormal returns (%), t-statistics in parentheses (Tables 4 and 5)

Window (months)	Fire sales	Unforced sales	Forced purchases
$[t-6, t]$	-10.07 (-3.68)	-12.21 (-7.48)	$+6.10$ (3.93)
$[t+1, t+12]$	$+6.13$ (2.47)	$+0.37$ (0.11)	-0.73 (-0.23)
$[t+1, t+18]$	$+9.65$ (3.47)	$+0.64$ (0.22)	-3.86 (-1.87)

- **Fire-sale stocks** fall about 15% before and during the event quarter, then **reverse** over the next 18 months (Table 4, Panel A; Figure 2).
- **Unforced sales** fall just as much but **do not reverse**—consistent with information, not price pressure (Table 4, Panel B; Figure 3).
- **Forced purchases** run up and later reverse (weaker; Table 5).
- Average forced seller sells at a **discount of about 5%** to fair value.

6.13 Coval-Stafford (2007): Trading on fire sales and implications

- **Liquidity provision:** long recent fire sales, short recent forced buys: CAPM alpha 0.97%/month ($t = 3.98$), four-factor 0.45% ($t = 2.00$) (Table 7, Panel C).
- **Front-running:** predict next quarter's flows from past flows and returns; long stocks expected to be bought, short stocks expected to be fire-sold: FF3 alpha **1.07%/month** ($t = 3.59$), four-factor 0.65% (Table 8, Panel C).
- **Implications:**
  - Even large, liquid stocks pay a large **price for immediacy**; demand curves are not flat.
  - Flow-driven trades **feed back** into fund performance (own and peers') (Edelen, 1999).
  - Flow-driven trading is a **non-fundamental shock to prices**—what corporate finance needs as an *instrument* for stock prices. That use is where the measure's weaknesses matter most.

6.13 Coval-Stafford (2007): Caveats of the *PRESSURE* measure

- **Caveat 1: actual trades are chosen.** *PRESSURE* uses the **realized** sales of outflow funds. Which positions to sell is a **decision**: the manager sells what she likes least (*information*) or what is most liquid (*selection*). Fire-sale stocks are not a random draw from the portfolio, so part of the return pattern may be information rather than pressure. The "unforced sales" comparison helps, but a stock-level *instrument* needs more.
- **Caveat 2: trades respond to prices.** Funds sell what has fallen and hold what is rising, and the measure is scaled by **volume**, another endogenous quantity. A measure built from realized trades and realized volume can be correlated with the very returns it is meant to explain.
- **Caveat 3: extreme-flow funds only.** Top/bottom flow deciles discard most of the variation in flows and make the measure sensitive to the cutoff.
- **Two follow-on papers** fix these in turn: Edmans, Goldstein, and Jiang (2012) replace actual trades with **hypothetical** ones; Wardlaw (2020) shows that the EGJ scaling still embeds the **realized return** and proposes a clean scaling.

6.13 Edmans, Goldstein, and Jiang (2012): First Fix—Hypothetical Trades

- **Idea:** assume a fund hit by a large outflow sells **pro rata**—every position in proportion to its weight—so the shock depends only on *pre-determined* holdings and *fund-level* flows, not on what the manager chooses to sell.
- **As defined in the paper (equations (1)–(2)):** $s_{i,j,t-1}$ shares of stock i held by fund j; PRC price; $VOL_{i,t}$ dollar volume in quarter t; $F_{j,t}$ flow as a fraction of fund assets; sum over funds with $F_{j,t} \leq -5\%$:

$$MFFlow_{i,t} = \sum_{j=1}^{m} \frac{F_{j,t} \times s_{i,j,t-1} \times PRC_{i,t-1}}{VOL_{i,t}}, \quad F_{j,t} = \frac{TA_{j,t} - TA_{j,t-1}(1 + R_{j,t})}{TA_{j,t-1}}$$

- **In words:** the numerator is the **dollar value of stock i that fund j would sell** to meet the outflow pro rata (= portfolio weight × dollar outflow); summed over outflow funds and divided by volume: hypothetical flow-driven selling relative to normal trading.
- **Use as an instrument:** $MFFlow$ predicts a transitory price drop; an inter-quartile increase in the instrumented valuation discount raises next year's takeover probability by **7 percentage points** (unconditional 6.2%).

6.13 Wardlaw (2020): Second Fix—What $MFFlow$ Actually Contains

- **The problem** (Wardlaw's decomposition). Write dollar volume as shares traded times price, $VOL_{i,t} = SHRVOL_{i,t} \times PRC_{i,t}$; then $MFFlow$ factors into three terms:

$$MFFlow_{i,t} = \underbrace{\sum_j F_{j,t} \times s_{i,j,t-1}}_{\text{flow-induced shares sold}} \times \underbrace{\frac{1}{SHRVOL_{i,t}}}_{\text{event-qtr share volume}} \times \underbrace{\frac{PRC_{i,t-1}}{PRC_{i,t}}}_{1/(1+R_{i,t})}$$

- The third term is the **inverse of the stock's own gross return** in the event quarter: a stock that fell gets a mechanically larger $MFFlow$. **The instrument contains the dependent variable.**
- The second term, **contemporaneous share volume**, is itself correlated with past and current returns and with news.
- By construction, extreme $MFFlow$ deciles are dominated by **small stocks**, where fund positions are large relative to volume.

6.13 Wardlaw (2020): The fix and the results

- **The fix:** drop the price term, and preferably the volume term too. Wardlaw's two measures scale the flow-induced share sales by shares **outstanding** ("flow-to-stock") or by share **volume** ("flow-to-volume"):

$$FIT_{i,t} = \frac{\sum_j F_{j,t} \times s_{i,j,t-1}}{SHROUT_{i,t-1}}, \qquad FIV_{i,t} = \frac{\sum_j F_{j,t} \times s_{i,j,t-1}}{SHRVOL_{i,t}}$$

  FIT contains nothing that depends on the event-quarter return; FIV still contains contemporaneous volume.

- **Results:** with the corrected measures, flow-induced selling produces only a **small decline** in the event quarter and **no subsequent reversal**; many results built on $MFFlow$ in the corporate finance literature do not survive. The price impact of hypothetical pro-rata selling is far weaker than that of *actual* fire sales in Coval–Stafford.

- **Lesson:** a measure of non-fundamental price movements must not contain the realized price itself; check every scaling variable.

6.14 Frazzini and Lamont (2008): Dumb Money

- **Question:** Coval–Stafford study *forced* trades over a quarter. Over longer horizons, do **retail flows across funds** act as a **sentiment** measure for individual stocks? Do stocks receiving flow-driven demand subsequently **underperform**, and do investors lose wealth by reallocating?
- **Idea:** when individuals move money from fund A to fund B, they implicitly demand B's stocks and sell A's. Aggregating across funds gives each stock's demand from *fund flows* rather than from managers' choices. "Smart money" (Gruber, 1996; Zheng, 2000) predicts *high* returns; the sentiment view predicts *low* returns.
- **Data:** Thomson/CDA holdings × CRSP fund flows, 1980–2003; flows inferred from TNA and returns as in Sirri–Tufano.
- **Contrast with Coval–Stafford:** CS: *extreme* flows, *actual* trades, *one quarter*, reversal within 18 months. FL: *all* flows, *hypothetical* proportional allocation, *three years*, returns over the following years. Both measure non-fundamental demand; they differ in horizon and in what counts as the shock.

6.14 Frazzini-Lamont (2008): The $FLOW$ measure (1/2)

Fund i, stock j (the paper's convention); $k = 12$ quarters. **Idea:** compare the sector's **actual** ownership of stock j with a **counterfactual** in which no money moved *across* funds since $t - k$.

- **Step 1—Fund flows** (eq. 1), net of mergers MGN:

$$F_{it} = TNA_{it} - (1 + R_{it})\, TNA_{i,t-1} - MGN_{it}$$

- **Step 2—Counterfactual fund size** (eqs. 2–3): from $t - k$ on, give each fund its **own returns** but only a **pro-rata** share (its size at $t - k$) of the sector's aggregate flow F_s^{Agg}, for $t - k \leq s \leq t$:

$$\hat{F}_{is} = \frac{TNA_{i,t-k}}{TNA_{t-k}^{Agg}} F_s^{Agg}, \qquad \widehat{TNA}_{is} = (1 + R_{is})\, \widehat{TNA}_{i,s-1} + \hat{F}_{is}$$

- **Step 3—Fund shares of the sector** (eqs. 4–5): actual vs. counterfactual,

$$x_{it} = TNA_{it}/TNA_t^{Agg}, \qquad \hat{x}_{it} = \widehat{TNA}_{it}/\widehat{TNA}_t^{Agg}$$

Funds that attracted **more than their pro-rata share** of flows have $x_{it} > \hat{x}_{it}$.

6.14 Frazzini-Lamont (2008): The $FLOW$ measure (2/2)

- **Step 4—Sector ownership of stock j** (eqs. 6–7): with w_{ijt} fund i's portfolio weight in stock j, add up each fund's dollar holding and scale by market cap:

$$z_{jt} = \frac{\sum_i x_{it}\, w_{ijt}\, TNA_t^{Agg}}{MKTCAP_{jt}}, \qquad \hat{z}_{jt} = \frac{\sum_i \hat{x}_{it}\, w_{ijt}\, TNA_t^{Agg}}{MKTCAP_{jt}}$$

- **Step 5—The measure** (eq. 8): actual minus counterfactual ownership,

$$FLOW_{jt} = z_{jt} - \hat{z}_{jt} = \frac{\sum_i [x_{it} - \hat{x}_{it}]\, w_{ijt}\, TNA_t^{Agg}}{MKTCAP_{jt}}$$

  the **excess dollars** that flowed into the funds holding stock j (beyond pro rata), weighted by how much of j each fund holds, as a fraction of the stock's market cap.

- **Interpretation:** high for stocks held by funds that have been **receiving money** (growth/tech funds, late 1990s): stock-level **retail sentiment** via fund choice. **Note:** scaled by market cap, no contemporaneous volume—FL anticipated Wardlaw.

6.14 Frazzini-Lamont (2008): High-flow stocks underperform for years

Monthly returns of high-minus-low $FLOW$ decile portfolios, 1983–2003 (Tables IV and VI); t-statistics in parentheses

Sorting horizon / adjustment	High − Low (%/month)	
3-year flow, equal-weighted, raw	−0.61 (−3.37)	about −8% per year
3-year flow, value-weighted, raw	−0.67 (−2.79)	
3-year flow, characteristic-adjusted	−0.22	size/BM/momentum
3-year flow, FF3 alpha	−0.45 (−2.95)	
5-year flow, equal-weighted	−0.32 (−2.37)	
3-month flow, equal-weighted	+0.11 (0.57)	no short-run smart money

- **Stocks that fund investors have been buying through their fund choices earn low subsequent returns**—negative in 17 of 21 years, though much of the cumulative effect comes after 1998 (Figure 2).
- **Related to value:** high-$FLOW$ stocks are growth stocks; characteristic adjustment removes most but not all of the effect. **Supply responds:** the highest-flow firms issue about 1% more equity per year (Table XII).

6.14 Frazzini-Lamont (2008): The cost to investors, and how this fits with Coval–Stafford

- **The "dumb money" effect:** actual sector return vs. the no-reallocation counterfactual: reallocation costs about **0.05% per month**, one-third to one-half of the sector's 1.4%/yr shortfall to the market (Table X).
- **Two horizons, one mechanism:** **quarters** (Coval–Stafford): forced trades move prices 5–10%, reverting within 18 months—**liquidity** pressure; **years** (Frazzini–Lamont): persistent flows into popular funds push up their stocks, which underperform for years—**sentiment** pressure. Lou (2012): flow-induced trading explains part of **momentum** and its reversal.
- **Reconciling dumb money with smart money** (Gruber, 1996; Zheng, 2000): smart money = funds receiving inflows do better over the **next quarter**. FL see a trace of it at 3 months (+0.11%/month, insignificant), but the sign **flips** at 1–3 years. Why: flows chase **past winners**, whose stocks carry **momentum** (Sapp and Tiwari, 2004)—the short-run gain is momentum, not information—while the long-run loss is the **reversal** of flow-driven price pressure. Smart vs. dumb is a matter of **horizon**; investors hold for years.

6.15 What Have We Learned?

- **The benchmark matters; the question decides the tool.** Holdings benchmarks have more power; factor models on net returns measure what investors get. Much apparent skill is an omitted factor, especially **momentum**.
- **Gross vs. net.** Stock picks roughly cover costs; investors' net alpha is about zero or negative (Wermers, 2000).
- **The Berk view.** Zero net alpha and no persistence are what skill looks like in equilibrium (Berk and Green, 2004); as **dollar value added**, skill is positive and persistent (Berk and van Binsbergen, 2015); flows reveal the **CAPM** as investors' yardstick.
- **Flows are an incentive contract.** Flows are **convex** in past performance (Sirri and Tufano, 1998): a call option; managers behind the market **shift risk** at year end (Chevalier and Ellison, 1997).
- **Flows move prices.** Forced trades create pressure that reverses (Coval and Stafford, 2007); clean measurement needs hypothetical trades and no embedded returns (Wardlaw, 2020); over years, flow-driven demand is "dumb money" (Frazzini and Lamont, 2008).

Reading List for Mutual Funds

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